A working model of behavior leakage: a constant write, gated by context similarity

June 2026

1 Setup

Leakage: a behavior trained into a model under one context (a system prompt, persona, or ICL prefix) also appears under other, untrained contexts. We measure it as the trained−base change in the behavior's expression under a bystander context, on-policy, with a behavior-appropriate metric: marker-token log-probability at the end of the bystander's own response (#502/#489), judge-scored agreement rates for sycophancy (#509), misaligned-and-coherent rates for emergent misalignment (#521), taught-fact emission for fact implants (#500).

We summarize a context c by an activation vector $v_c \in \mathbb{R}^d$ and a behavior by the *behavior state* $v_b(c) = f(c)$, the activations where the behavior is expressed, with f the model's unknown context-to-behavior map. Each behavior also has a *read-out direction* r_b whose projection tracks expression (the unembedding row for the marker; a probe or contrastive direction otherwise). The constructions of v_c and v_b (layer, position, aggregation) are swept, not fixed; the experiments in §4 search over them.

2 The simplest model

Assumptions. Each comes with the relaxation that applies when it fails; the formula survives every relaxation in a weaker form.

A1 (behaviors are linear directions).
Adding a single residual-stream direction elicits a behavior [3, 4]; projections onto single directions track traits before generation [5]; emergent misalignment is predicted, controlled, and ablated through one direction [6, 7]. *Relaxation.* The *write* need not copy the behavior direction: it can be a router, a query shift that recruits pre-existing attention circuitry which fetches the behavior from the context's own values [8]. The sharpest evidence is a rank-1 EM adapter whose B vector has cosine 0.04 to the misalignment direction at its own layer while 98% of its effect ablates along that direction downstream [7]. The behavior stays one direction at read-out, but prediction P1 below can fail by mechanism when fetched content differs across contexts.

A2 (edits are read/write pairs).
Weight updates read input directions and write output directions, with the rank-one outer product as the atomic unit [1]; low-rank adapters make this explicit [16, 2]. Narrow fine-tunes act as a near-constant steering write [8, 9], and the parameterization is irrelevant: at matched implant strength, LoRA and full fine-tuning leak identically (gap +0.00 nat, #514). *Extension.* The edit need not come from training: a transformer block provably converts an ICL prefix into a transient rank-one weight patch [17, 18], so an in-context prefix

performs the same write $\times$ gate operation for the duration of the prompt, and the gate already ranks leakage from ICL sources (#489). Full ICL=fine-tuning equivalence remains open for pretrained models [19, 20, 21], but in-context behavior carries predictive signal about fine-tuning outcomes: ICL recovers $> 85\%$ of fine-tuning's corrections in-distribution [19], in-context inferences anticipate and repair fine-tuning's generalization failures [22], and demonstration effects compress to a single latent direction [23]. Hence prediction P7 below.

A3 (linearity, the extra assumption).
At one layer, $v_b \approx M v_c$, and the edit acts at that layer. *Relaxation.* For a small update, the change at any input is first-order: $\Delta f(c) \approx K(c, c') \times$ residual, with K the empirical tangent kernel [14], a regime measured to hold for language-model fine-tuning in several settings [15]. The formula survives with $g(c) = K(c, c')/K(c', c')$, where K is the cross-kernel between the read-out gradient at c and the training-loss gradient at c' (symmetric only for read-out-aligned losses such as marker-only). The dot product of equation (1) is the linear-kernel case, and the practical ladder runs cosine $\rightarrow$ distributional comparisons (Gaussian-KL, MMD) $\rightarrow$ the exact cross-kernel via backprop; gains over cosine have so far been marginal (0.79 vs 0.76 marker; 0.50 vs 0.49 sycophancy). The gate can also stop being a similarity altogether: under the routing picture of A1's relaxation it is attention-shaped, near-binary for discrete triggers, and estimable from samples rather than computable from cosines.

A4 (minimal edit).
Training implements the smallest weight change that fits the new pair: the rank-one key$\rightarrow$value write used by linear associative memories and ROME [10, 11, 12]. *Relaxation.* The write becomes a low-rank subspace, $\Delta v_b(c) = U\,\mathrm{diag}(s)\,V^\top v_c$; prediction P1 below weakens to "shifts span r dimensions," and the SVD already in use measures r. With multiple training pairs the edit solves a kernel system over all pairs; contrastive negatives enter as off-diagonal terms that subtract the common mode (persona contexts sit near cosine 0.92, which is why positive-only implants leak near-uniformly while contrastive ones localize: $\sim$99.6% source vs $\sim$11.7% bystander, #247/#329). Early stopping spectrally filters the solution toward raw similarity, so the time-indexed predictor interpolates between raw and Gram-corrected similarity; neither endpoint has been run.

From assumptions to formula. Fine-tune context c' from behavior state $v_{b'} = f(c')$ to an achieved state $v_{b''}$ (under early stopping the achieved state undershoots the target, so the write is measured, not assumed). A4 with A3 gives the edit: minimizing $\|\Delta M\|_F$ subject to $(M+\Delta M)\, v_{c'} = v_{b''}$ yields

$$M' = M + \frac{(v_{b''} - v_{b'})\, v_{c'}^\top}{\|v_{c'}\|^2} \qquad \Longrightarrow \qquad \Delta v_b(c) \;=\; \underbrace{(v_{b''} - v_{b'})}_{\text{write: } \textit{what} \text{ leaks}} \;\cdot\; \underbrace{\frac{\langle v_{c'}, v_c\rangle}{\|v_{c'}\|^2}}_{\text{gate: } \textit{who} \text{ leaks}} \;. \tag{1}$$

Each assumption does one job: A4 makes the edit rank-one (one write direction, one key), A3 makes the gate the activation inner product, A2 makes the same structure visible in the weights ($\Delta W \approx \text{write} \otimes \text{key}$), and A1 makes the write measurable, since the observed change is one slope per behavior times the gate: $\Delta z_b(c) = \langle r_b,\, v_{b''} - v_{b'}\rangle\, g(c)$. The softmax compresses log-probabilities near saturation and can reorder contexts there, which is why we store logits alongside log-probabilities.

3 Predictions

P0 (weights decompose).
The top singular vectors of the trained ΔW are the key and the write. Under contrastive training the key moves from raw $v_{c'}$ toward a source-minus-negatives contrast direction as dose grows.

P1 (one write direction).
Per-bystander activation shifts at the read-out all point along $(v_{b''} - v_{b'})$; an SVD of the shift matrix is near rank-one with a top component matching the behavior delta.

P2 (the gate is base-model context similarity).
Some fixed kernel on base-model context summaries sets who leaks, and it transfers across held-out panels. On the linear rung it is the dot product, hence asymmetric: $c' \to c$ scales as $\cos\theta \cdot \|v_c\|/\|v_{c'}\|$.

P3 (the behavior's prior matters, through three channels).
Source side: the write scales with the residual $\|v_{b''} - v_{b'}\|$, so a source that already performs the behavior produces little leakage despite high similarity. Bystander side: the prior enters through read-out compression (measurement) and possibly through the gate itself (mechanism); the two separate in logit space, but only with the exact gate rather than a proxy.

P3′ (the source prior shapes the write's direction, not just its size).
If the source already carries the behavior's shared component, little of it needs writing: the residual concentrates in source-specific subspace, so the source prior predicts the write's *common-mode fraction* (its projection onto the shared / mean-bystander shift direction), not its norm. This is the configuration #541 observed — the high-prior teachers express the implant *most* on themselves (97/94% self-assertion vs the leaky low-prior teacher's 72%) while leaking least ("not weaker, narrower") — which the magnitude-only reading of P3 cannot produce: a uniformly small write would weaken expression everywhere, including source-self.

P4 (additivity).
Two implants trained jointly leak like the sum of their singletons, with deviations set by the sources' mutual similarity (the activation-space analogue of task arithmetic [13]).

P5 (layer locality).
The gate predicts best at the layers where the behavior is computed; scoring this needs an independent localization (patching or probing), since scoring by predictor quality is circular.

P6 (cross-behavior transfer).
The same write moves other behaviors in proportion to its projection onto their read-out directions, extending the rule from context generalization to behavior generalization.

P7 (ICL proxy).
The base model's in-context behavior on the training mix forward-predicts the post-training leakage map (see A2's extension for why).

No single miss refutes the model, since each assumption carries a relaxation (§2); the conjunction does. With the gate computed exactly and the write measured on a rig-clean implant, a failure of write $\times$ gate to rank leakage on a held-out panel leaves no version standing.

4 Experiments

	Experiment	Status
P0	SVD stored adapters; compare singular vectors to keys and measured writes; cross-compare [8]'s per-trigger backdoor vectors	Not yet run; free on existing artifacts.
P1	SVD per-context activation shifts, marker implant vs EM SFT (#521, controls #551)	Split: EM shifts all 14 contexts along one seed-stable direction (cos 0.96–0.98); the marker implant does not, and its trained context is *least* aligned. Benign SFT also gives one direction (#552, running); positive-only retrain tests the rig confound (#561, running).
P2	Correlate base-model similarity with measured leakage across source×bystander panels (#502, #489, #509)	Similarity ranks marker leakage ($\lvert\rho\rvert \approx 0.76$–$0.79$, layers 19–24) and extends to ICL/multi-turn contexts, but only ordinally (out-of-fold $R^2 \approx 0$, single-seed); structure does not transfer across panels (#553); the fact arm reversed sign ($\lvert\rho\rvert = 0.49$) while the base prior predicted correctly (+0.27, #500), an in-regime miss.
P3	Instructed-source break case; regress logit shifts on gate proxy + bystander base logit (#500, #531, #532)	The bystander's own prior is the most consistently supported predictor and beats geometry on instruction contexts (#532); the measurement-vs-mechanism split awaits the exact gate.
P3′	Decompose measured writes: project per-context activation shifts onto shared (mean-bystander) vs source-specific components; regress the common-mode fraction on source prior (#541's nine fact adapters span teacher prior; #518's refusal/EM adapters extend across behaviors)	Not yet run; activation extraction on existing adapters only, no new training.
P4	Joint two-source training vs singleton sum (#520, #527, #538)	Not yet diagnostic: singleton shifts are themselves near rank-one, so the additivity cosine (0.99) is trivially high at every tested dose.
P5	Layer sweep of predictor quality per behavior + independent localization	Marker predicts best late (19–24), sycophancy early (1–8); consistent but unconfirmed without the localization step.
P6	Behavior-generalization testbed (#545); context testbed (#537); formula-as-predictor (#510)	#537 running; #545, #510 queued.
P7	ICL-vs-FT equivalence, persona-framed (#491): same K examples in-context vs fine-tuned; ICL-conditioned base responses as a leakage predictor	Queued; the similarity gate already extends to ICL source contexts (#489).

The open headline experiment is a dose titration: implant at a non-saturated anchor (#448), measure expression change, activation shift, and gate proxy per held-out context, on-policy (#432→#456), at several doses across the 10–25-step cliff (#139), with at least 3 seeds. At low dose the shift matrix should be near rank-one with similarity-graded magnitude; across the cliff the direction should rotate and the prediction collapse. The P1 split already complicates the low-dose half, so the titration doubles as its diagnosis.

What cannot be relaxed. Every version assumes the update is small in weight/feature space, though not in behavior: read-outs amplify, so a lazy edit can flip emission discretely or induce EM through nine rank-1 adapters [7], and tripling implant depth left the edit geometry unchanged (#538). If training moves the features themselves (the rich regime), no base-model predictor can work, and the historical EM predictor failure looks like that regime ($\rho = 0.09$ prose-only #458; assistant-axis rotation 38–53°, discrimination collapse, #125). The current tension: the marker arm passes the gate test but fails the one-direction test, while EM does the reverse; #552 and #561 decide whether the rig or the regime explains it. The durable question under all versions is the same: how far is fine-tuning generalization predictable from base-model activation geometry, before training?

References

[1] N. Elhage et al. A mathematical framework for transformer circuits. *Anthropic (transformer-circuits.pub)*, 2021.

[2] J. Ward et al. Rank-1 LoRAs encode interpretable reasoning signals. arXiv:2511.06739, 2025.

[3] A. Turner et al. Activation addition: steering language models without optimization. arXiv:2308.10248, 2023.

[4] N. Panickssery et al. Steering Llama 2 via contrastive activation addition. arXiv:2312.06681, 2023.

[5] R. Chen, A. Arditi, H. Sleight, O. Evans, J. Lindsey. Persona vectors: monitoring and controlling character traits in language models. arXiv:2507.21509, 2025.

[6] M. Wang et al. Persona features control emergent misalignment. arXiv:2506.19823, 2025.

[7] A. Soligo, E. Turner, S. Rajamanoharan, N. Nanda. Convergent linear representations of emergent misalignment. arXiv:2506.11618, 2025.

[8] J. Engels et al. Simple mechanistic explanations for out-of-context reasoning. arXiv:2507.08218, 2025.

[9] J. Minder et al. Narrow finetuning leaves clearly readable traces. arXiv:2510.13900, 2025.

[10] T. Kohonen. Correlation matrix memories. *IEEE Trans. Computers*, 1972.

[11] J. A. Anderson. A simple neural network generating an interactive memory. *Mathematical Biosciences*, 1972.

[12] K. Meng, D. Bau, A. Andonian, Y. Belinkov. Locating and editing factual associations in GPT. *NeurIPS*, 2022.

[13] G. Ilharco et al. Editing models with task arithmetic. *ICLR*, 2023.

[14] A. Jacot, F. Gabriel, C. Hongler. Neural tangent kernel: convergence and generalization in neural networks. *NeurIPS*, 2018.

[15] S. Malladi, A. Wettig, D. Yu, D. Chen, S. Arora. A kernel-based view of language model fine-tuning. *ICML*, 2023.

[16] E. Hu et al. LoRA: low-rank adaptation of large language models. *ICLR*, 2022.

[17] B. Dherin et al. Learning without training: the implicit dynamics of in-context learning. arXiv:2507.16003, 2025.

[18] Goldwaser et al. arXiv:2511.17864, 2025 (extends [17] to gated, bias-free, RMSNorm transformer blocks).

[19] D. Dai et al. Why can GPT learn in-context? Language models implicitly perform gradient descent as meta-optimizers. arXiv:2212.10559, 2022.

[20] L. Shen, A. Mishra, D. Khashabi. Do pretrained transformers really learn in-context by gradient descent? arXiv:2310.08540, 2023 (ICML 2024).

[21] G. Deutch et al. In-context learning and gradient descent revisited. arXiv:2311.07772, 2023 (NAACL 2024).

[22] A. Lampinen et al. On the generalization of language models from in-context learning and finetuning: a controlled study. arXiv:2505.00661, 2025.

[23] S. Liu et al. In-context vectors: making in-context learning more effective and controllable through latent space steering. arXiv:2311.06668, 2023 (ICML 2024).